Answers – Error Detection Methods

Question 1 – Two-Dimensional Parity Check

Datawords: 1010, 0111, 1100, 1001. Even parity is used.

a) Row parity bits are 0, 1, 0, and 1. Column parity bits are 0, 1, 1, and 1. The corner bit is 0.

```
1 0 1 0 | 0
0 1 1 1 | 1
1 1 0 0 | 0
1 0 0 1 | 1
----------------
0 1 1 1 | 0
```

Complete transmitted block:

```
10100
01111
11000
10011
01110
```

b) Received block:

```
10100
01111
11000
10011
10110
```

Row checks: rows 1–4 have even parity, but the parity row has 3 ones (odd). Column checks: columns 1 and 2 are incorrect. Therefore, the received message is NOT accurate; an error is detected. With multiple inconsistent parity checks, the block cannot be accepted as error-free.

Question 2 – Simple Parity Check

a) Add one parity bit so that the total number of 1s is even.

```
1010101 → 10101011

1111000 → 11110001

0000000 → 00000000
```

b) Receiver check:

```
10101010 → INVALID (3 ones: odd)

11110001 → VALID (5 ones? Actually 11110001 has five 1s, so INVALID)

00000001 → INVALID (1 one: odd)
```

A simple parity check detects an odd number of bit errors, but it cannot locate or correct the erroneous bit. Therefore, the receiver cannot correct these errors.

Question 3 – Internet Checksum

Sender words:

```
1100110011001100

1010101010101010
```

Add using one's-complement addition:

```
1100110011001100

+1010101010101010

----------------

1 0111011101110110


End-around carry:

0111011101110110 + 1 = 0111011101110111
```

Checksum = one's complement of the sum:

```
1000100010001000
```

Transmitted packet:

```
1100110011001100

1010101010101010

1000100010001000
```

Receiver calculation:

```
1100110011001100

+1010101010101010

+1000100010001000

----------------

1111111111111111
```

The final result is all 1s, so the packet is accepted as error-free.

Question 4 – Checksum Error Detection

a) Receiver adds the two words:

```
1111000011110000

+0000111100001111

----------------

1111111111111111
```

The correct checksum should therefore be 0000000000000000. Since the received checksum is 0000000000000000, the receiver obtains:

```
1111111111111111

+0000000000000000

----------------

1111111111111111
```

All 1s means the packet is accepted as error-free.

b) If the received checksum is 1111111111111111:

```
1111111111111111 + 1111111111111111

= 1 1111111111111110

End-around carry → 1111111111111111
```

The final result is still all 1s, so the receiver would accept the packet. This illustrates that a checksum cannot detect every possible error pattern.

Question 5 – CRC

Generator = 1011. Dataword = 11010011. Append three zeros:

```
11010011000
```

Modulo-2 division by 1011 gives remainder 011:

```
11010011000 ÷ 1011 → remainder 011
```

Transmitted codeword = data + CRC:

```
11010011 011 = 11010011011
```

b) Receiver gets 11010011101:

```
11010011101 ÷ 1011 → remainder 110
```

Remainder ≠ 000, so the frame is REJECTED; an error is detected.

Question 6 – CRC Error Detection

Generator = 1101. The transmitted codeword is 10101010101.

a) Received codeword = 10101010111:

```
10101010111 ÷ 1101 → remainder 001
```

Remainder ≠ 000, so an error is detected and the frame is rejected.

b) Received codeword = 10101010101:

```
10101010101 ÷ 1101 → remainder 000
```

Remainder = 000, so the frame is accepted.

Question 7 – CRC Full Sender and Receiver

Data = 101110. Generator polynomial $x^3 + x + 1$ corresponds to 1011.

a) Append three zeros:

```
101110000
```

Modulo-2 division:

```
101110000 ÷ 1011 → remainder 100
```

CRC = 100. Transmitted frame:

```
101110100
```

b) Receiver gets 101110101:

```
101110101 ÷ 1011 → remainder 001
```

Remainder ≠ 000, so the frame is rejected.

c) Receiver gets 101110100:

```
101110100 ÷ 1011 → remainder 000
```

Remainder = 000, so the frame is accepted.

Question 8 – Two-Dimensional Parity with Burst Error

Data:

```
111

000

101
```

a) Row parity bits: 1, 0, 0. Column parity bits: 0, 1, 0. Corner bit = 1.

Complete transmitted block:

```
1 1 1 | 1

0 0 0 | 0

1 0 1 | 0

---------

0 1 0 | 1
```

b) Received block:

```
1 1 1 | 1

0 0 0 | 0

1 0 1 | 0

---------

0 1 0 | 0
```

The data rows and columns remain even, but the parity row is incorrect: it contains only one 1 instead of two. Therefore, an error is detected in the parity information (the corner bit). The received block is not accepted as error-free.

Question 9 – Checksum with Multiple Words

Use 8-bit one's-complement addition.

```
10101010
01010101
--------
11111111


11111111
+11110000
--------
1 11101111


End-around carry:
11101111 + 1 = 11110000


11110000
+00001111
--------
11111111
```

Sum = 11111111. Therefore, the 8-bit checksum is:

```
00000000
```

b) Receiver verification:

```
10101010
+01010101
+11110000
```

```
+00001111

+00000000

-----------

11111111
```

The result is 11111111, so the packet is accepted as error-free.

Question 10 – Compare the Three Methods

Dataword = 1101.

a) Even parity: 1101 contains three 1s, so the parity bit must be 1.

```
Codeword = 11011
```

Receiver counts all 1s. If the total is even, the codeword passes the parity check. Here, 11011 has four 1s, so it is valid.

b) CRC with generator 1011. Append three zeros:

```
1101000 ÷ 1011 → remainder 111
```

CRC = 111. Transmitted frame:

```
1101 111 = 1101111
```

Receiver division:

```
1101111 ÷ 1011 → remainder 000
```

Remainder 000 means the frame is accepted.

c) Checksum: split 1101 into two 2-bit words: 11 and 01.

```
11 + 01 = 100
```

Using 2-bit one's-complement arithmetic, wrap the carry:

```
00 + 1 = 01
```

Checksum = one's complement of 01:

```
10
```

Receiver verification:

```
11 + 01 + 10 = 100

Wrap carry: 00 + 1 = 01

One's complement of 01 = 10
```

Equivalently, adding the data and checksum with one's-complement arithmetic gives all 1s (11), so the packet is accepted.

Important Notes

1. Simple parity can detect many errors but cannot locate/correct an error.
2. A CRC frame is accepted only when the receiver's division gives a zero remainder.
3. An Internet checksum packet is accepted when the one's-complement sum of all words, including the checksum, is all 1s.
4. In Question 4(b), the particular error pattern is not detected by the checksum because the final one's-complement sum remains all 1s.